Lucas Newcomb

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Education

University of Maryland, College Park — College Park, MD
B.S. Computer Engineering (intended) — *Aug. 2026 – May 2029*

Rochester Institute of Technology — Rochester, NY
Computer Engineering, BS/MS accelerated program; GPA: 3.98/4.00, Dean's List — *Aug. 2025 – May 2026*

- **Coursework (A in each):** Calculus I–II, University Physics I, Computer Science I–II, Digital System Design I

Technical Experience

Flight Software Responsible Engineer, Data Downlink — Oct. 2026 – Present
UMD SatDev — TERP RAPTOR — *College Park, MD*

- Own the flight software side of payload data downlink for a student CubeSat mission to the 2029 Apophis flyby.
- Building the mission data budget and onboard storage design, sizing each phase's data against storage and downlink.
- Researching and selecting the lossless compression algorithm for asteroid imagery, run on a single onboard MCU.

Embedded Software Team Member — Aug. 2025 – May 2026
RIT Space Exploration (SPEX) — Rover Team — *Rochester, NY*

- Wrote the rover's CAN messaging library — the shared communication layer between the Teensy 4.1 controllers running the arm, chassis, and science subsystems — using dynamic dispatch to route incoming frames to registered per-message handlers, so subsystem code declares what it listens for instead of parsing the bus itself.
- Authored much of the rover's microcontroller firmware in C++ on PlatformIO; the arm and life-detection subsystems both ran on the library.
- Helped build a software simulator that mocks the rover's subsystems, letting the team exercise and test subsystem code without the physical rover.

Student Intern — July 2024 – Aug. 2024
U.S. Army Research Laboratory — *Adelphi, MD*

- Developed a prototype grant proposal application on ServiceNow, building the workflow from submission through review.
- Worked within an agile software development team alongside full-time staff.

Projects

CUBERT — Flatsat Flight Software & Ground Station | *Rust, RTIC, STM32F446, Python* — 2026 – Present

- Built end-to-end flight software for a flatsat: `no_std` Rust firmware on an STM32F446 that samples sensors on a periodic schedule, sequences behavior through a state machine, and detects and recovers from faults at runtime.
- Used RTIC for deterministic, priority-based task scheduling with no dynamic allocation, and framed downlinked telemetry over serial with COBS so packet boundaries survive a noisy link.
- Wrote the matching ground station — a Python web dashboard that decodes the live telemetry stream and renders it as time-series graphs, current values, and subsystem status indicators.

LUNA — 3D N-Body Simulator | *C++, OpenGL* — 2025 – Present

- Built a real-time gravitational N-body simulator with two interchangeable solvers — a Barnes–Hut octree at $O(n \log n)$ and direct summation at $O(n^2)$ as an accuracy reference — integrated with a symplectic leapfrog scheme that keeps orbital energy stable over long runs instead of drifting.
- Multithreaded the force computation to sustain ˜160 FPS at 10,000 bodies while remaining interactive at 100,000; renders live in OpenGL and runs headless for batch jobs.

Additional Experience

Robotics Instructor — June 2026 – Aug. 2026
Unity Active — *Northern Virginia*

- Taught robotics to school-age students with no prior technical background.

Team Member, promoted to Shift Manager — Jan. 2025 – Aug. 2026
Papa John's — *Burke, VA*

- Worked the make line and register through high-volume shifts; promoted to shift manager in summer 2026.

Technical Skills

Languages: C, C++, Rust, Python
Embedded: STM32, Teensy, CAN, I2C, SPI, UART
Systems & Tools: Linux, Git, CMake/Ninja, PlatformIO, OpenGL, clang-tidy, cppcheck, ServiceNow